VPN / Transit Gateway / VPC Peering — Console Dashboard Wise Important Interview Questions & Answers- AWS Site-to-Site VPN

VPC
│
├— Virtual Private Gateways
├— Customer Gateways
├— Site-to-Site VPN Connections
└— Transit Gateways

1. What is AWS Site-to-Site VPN?
Answer:
AWS Site-to-Site VPN creates an encrypted connection between our on-premises network and an AWS VPC.
It uses IPsec tunnels to securely transfer traffic between the two networks.
We commonly use it when a company wants private connectivity between its data center and AWS.
Example:
Suppose our company has a database in its office and an application in AWS.
We create a Site-to-Site VPN so the AWS application can securely communicate with the office network.
The communication travels through encrypted VPN tunnels.

On-Premises
│
│ Encrypted VPN
↓
AWS VPN
│
↓
VPC
↓
EC2

Simple ga gurthupettuko:
Site-to-Site VPN = Secure connection between On-Premises and AWS

---

2. What is a Customer Gateway?
Answer:
A Customer Gateway represents our on-premises VPN device or firewall in AWS.
When creating it, we provide information such as the device's public IP address and routing details.
AWS uses this information to establish the VPN connection with our on-premises network.
Example:
Suppose our company has a firewall with a public IP address.
We create a Customer Gateway in AWS using that public IP.
AWS then uses it while creating the Site-to-Site VPN connection.

On-Premises Firewall
│
↓
Customer Gateway
│
↓
AWS VPN

Simple ga gurthupettuko:
Customer Gateway = Our side of the VPN

---

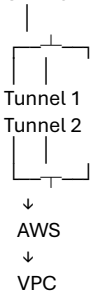
3. What is a Virtual Private Gateway?
Answer:
A Virtual Private Gateway, or VGW, is a VPN endpoint on the AWS side of a VPC.
We attach the Virtual Private Gateway to the VPC that needs connectivity with the on-premises network.
It works together with the Customer Gateway to establish Site-to-Site VPN connectivity.
Example:
Suppose our production VPC needs to connect to the company data center.
We attach a Virtual Private Gateway to the VPC and connect it with the Customer Gateway through VPN.

On-Premises
│
Customer Gateway
│
↓
VPN
│
↓
Virtual Private Gateway
│
↓
VPC

Simple ga gurthupettuko:
VGW = AWS side of traditional VPN

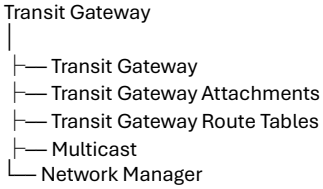
---

4. What is a Site-to-Site VPN Connection?
Answer:
A Site-to-Site VPN Connection is the actual VPN connection between our on-premises network and AWS.
It normally provides two VPN tunnels for high availability.
The tunnels use encryption to securely transfer traffic between the networks.
Example:
Suppose our company connects its data center to an AWS VPC.
We create a VPN connection between the Customer Gateway and AWS VPN endpoint.
If one tunnel has a problem, the other tunnel can provide connectivity.

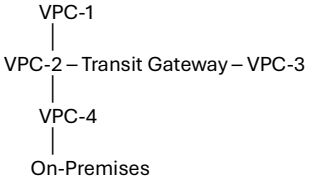


Simple ga gurthupettuko:
VPN Connection = Actual encrypted connection

AWS Transit Gateway



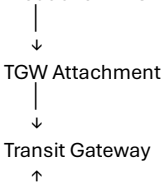
5. What is AWS Transit Gateway?
Answer:
AWS Transit Gateway is a central network hub used to connect multiple VPCs and on-premises networks.
Instead of creating separate connections between every VPC, we connect the VPCs to the Transit Gateway.
This makes large network architectures easier to manage.
Example:
Suppose a company has 10 VPCs that need to communicate with each other.
Instead of creating many individual connections, we attach the VPCs to one Transit Gateway.
The Transit Gateway acts as a central networking hub.



Simple ga gurthupettuko:
Transit Gateway = Central hub for many VPCs

---

6. What is a Transit Gateway Attachment?
Answer:
A Transit Gateway Attachment connects a network resource to a Transit Gateway.
We can attach VPCs, VPN connections, and other supported network resources.
After the attachment is created, routing determines how traffic moves through the Transit Gateway.
Example:
Suppose we have a Production VPC and a Development VPC.
We create attachments for both VPCs to the Transit Gateway.
Now the Transit Gateway can route traffic between them based on its route tables.

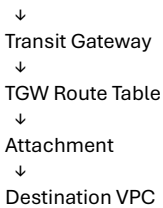




Simple ga gurthupettuko:
Attachment = Connect a network to Transit Gateway

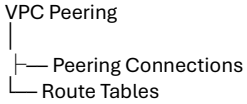
---

7. What is a Transit Gateway Route Table?
Answer:
A Transit Gateway Route Table controls where traffic should go after reaching the Transit Gateway.
It contains routes that point traffic toward the appropriate Transit Gateway attachment.
We can use different route tables to isolate or control traffic between networks.
Example:
Suppose Production and Development VPCs are connected to the same Transit Gateway.
We can configure routes so Production can access shared services while Development cannot access Production.
This gives us better control over network communication.

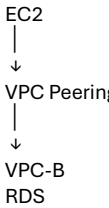


Simple ga gurthupettuko:
TGW Route Table = Decide where TGW traffic goes

VPC Peering



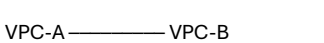
8. What is VPC Peering?
Answer:
VPC Peering creates a private network connection between two VPCs.
The VPCs can communicate using private IP addresses without sending traffic through the public internet.
After creating peering, we must add the required routes in the VPC route tables.
Example:
Suppose our application is running in one VPC and the database is running in another VPC.
We create VPC Peering between them and add routes for the required CIDR ranges.
The application can then communicate with the database using private IPs.



Simple ga gurthupettuko:
VPC Peering = Private connection between two VPCs

---

9. What is the difference between VPC Peering and Transit Gateway?
Answer:
VPC Peering directly connects one VPC to another VPC.
Transit Gateway provides a central hub through which many VPCs and networks can connect.
For a small number of VPCs, peering may be simple, while Transit Gateway is useful for larger network environments.
Example:
Suppose we have only two VPCs that need communication.
VPC Peering can be a simple solution.
If we have many VPCs and on-premises networks, Transit Gateway provides centralized connectivity.
VPC Peering:



Transit Gateway:



VPC-B ─┼─ Transit Gateway ─ On-Premises
VPC-C ─┘
Simple ga gurthupettuko:
Peering = VPC-to-VPC
Transit Gateway = Many networks through one hub

---

10. What should we check in the AWS Console when troubleshooting VPN/Peering/TGW?
Answer:
First, we check whether the VPN, peering connection, or Transit Gateway attachment is in the correct state.
Then we check route tables, CIDR ranges, security groups, and network ACLs.
We also verify that the return route exists so that traffic can travel back to the source.
Example:
Suppose EC2 in VPC-A cannot reach EC2 in VPC-B through peering.
We check the peering status and both VPC route tables.
Then we verify security groups and NACLs allow the required traffic.
EC2-A
↓
Route Table
↓
Peering / TGW
↓
Route Table
↓
Security Group / NACL
↓
EC2-B
Simple ga gurthupettuko:
Networking Troubleshooting = Status → Routes → SG/NACL → Return Route